MILITARY SPECIFICATION

GENERATOR, DIRECT CURRENT: G-76( )/G WITH ACCESSORIES

This specification is approved for use by USA Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements for Generator, Direct Current, G-76( )/G, a portable, hand-cranked electrical generator with accessories to provide power to military radios, coding equipment, rechargeable batteries, and other electronic equipment in the field.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

* MIL-P-116 Preservation-Packaging, Method of
MIL-P-11268 Parts, Materials and Processes Used in Electronic Communication Equipment
MIL-M-13231 Marking of Electronic Items
* MIL-F-14072 Finishes for Ground Electronic Equipment
MIL-L-81352 Lacquer, Acrylic (For Naval Weapons Systems)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Electronics Research and Development Command, Harry Diamond Laboratories, ATTN: DELHD-D-OE-PS, 2800 Powder Mill Road, Adelphi, MD 20783, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6115

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STANDARDS

Federal

FED-STD-595 Colors

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 Marking for Shipment and Storage
MIL-STD-252 Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454 Standard General Requirements for Electronic Equipmer
MIL-STD-461 Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462 Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781 Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution
MIL-STD-810 Environmental Test Methods

DRAWINGS

USA ERADCOM Harry Diamond Laboratories (FSCM 19202)

DL 11737694 Data List for Generator, Direct Current: G-76( )/G with Accessories
EDL 11737694 Equipment Data List for Generator, Direct Current: G-76( )/G with Accessories

USA CERCOM (FSCM 80063)

SPI 1G0061 Special Packaging Instructions for G-76( )/G

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Samples. The contractor is responsible for all requirements for providing and testing of samples as prescribed herein. When this specification or any portion thereof is assigned to a subcontractor, it shall be the prime contractor's responsibility to insure that sample requirements are met. The prime contractor shall notify the Government at the time of

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assignment to a subcontractor whether the testing will be performed by the prime or by the subcontractor, as well as the method of control by the prime contractor.

3.1.1 First article sample. Prior to the start of regular production, except when a new contract follows within 90 days production of acceptable material of identical design, the contractor shall manufacture and submit a first article sample of * twenty units (Hand Crank Generator Assembly - 11733326), using as far as possible the methods and processes proposed for quantity production. When processes differ from the production processes, they shall be basically equivalent and the difference shall be fully described in documentation submitted with the first article sample. The sample shall conform to the requirements of this specification. Additional samples that may be required because of failure of the sample to meet the requirements of this specification shall be supplied by the contractor at his expense. Prior to approval of the first article sample, acquisition of parts and materials or initiation of production will be at the sole risk of the contractor and the Government will not proceed with inspection for acceptance of production lots until this approval has been obtained.

3.1.2 Supplemental samples. At the Government's discretion, a supplemental sample may be required whenever there is a break in production continuity of 90 days or more, or a change in design, material, or process that may affect safety, operability, or reliability as defined by the requirements. In all cases, such changes will require prior approval of the procuring activity.

3.1.3 Comparison samples. One unit from each lot shall be shipped within two working days of lot acceptance to the technical agency designated by the procuring activity. (See 4.4.1 and 6.1.3.) This unit shall be drawn from the lot sample.

3.2 Construction. Construction shall be in accordance with DL 11737694 and the following.

3.2.1 Materials. Materials shall be those specified by the applicable drawings.

3.2.1.1 Harmonic drive assembly. The harmonic drive assembly (11733294) shall be thoroughly cleaned and lubricated using lubricant (11733314) prior to incorporation into the * electromechanical assembly (11733285).

3.2.2 Accessories. Accessories shall be required as specified in the contract. Accessories include:

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*	11738242	Accessory Kit, Field Radio Equipment, MK-2120/G
*	11738243	Accessory Kit, Permissive Action Link, MK-2121/G
*	11733296	Cable Assembly, Special Purpose, Electrical, CX-13175/G
*	11733297	Cable Assembly, Special Purpose, Electrical, CX-13176/G
*	11733365	Bag, Carrying
*	11733985	Adapter, Connector, MX-2122/G
*	11738274	Cover Assembly, Plate, Nonskid

3.2.3 Dimensions and technical notes.

3.2.3.1 Listed. Those dimensions and technical requirements listed in the Classification of Defects (4.6) are mandatory.

3.2.3.2 Unlisted. The contractor may request deviations from characteristics shown on the drawings, but not listed in the classification of defects, for the purpose of adapting the item to established manufacturing practices. Such requests must be accompanied by evidence that the deviation does not affect the design and that all performance requirements will be met. If the Government confirms the contention of the contractor, the deviation will be approved for the duration of the contract. In case of dispute, the characteristics of the drawings shall apply. Approval of a deviation under provision of this paragraph does not relieve the contractor from establishing and maintaining an adequate quality assurance program as elsewhere required. Provisions of this paragraph shall not be used to obtain approval for use of discrepant materials (i.e., nonconforming material produced before approval is obtained), or for design changes which should be requested in accordance with change provisions of the contract document. It is expected that the above deviations will normally be accomplished prior to the construction of the First Article Sample (FAS) and be reflected therein.

3.2.3.3 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment (See 6.3.1). Individual items shall not be handpicked for fit or performance. (Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.)

3.2.4 Protective finish. The surface of the unit shall be treated in accordance with MIL-F-14072, and painted with material * meeting MIL-L-81352. Exposed aluminum surfaces shall be prepared * per Finish P213.1 of MIL-F-14072. Exposed steel surfaces, such * as screws, fasteners, and chains, shall be prepared per Finish * P211.1 of MIL-F-14072. Exposed stainless steel surfaces shall be * passivated per Specification QQ-P-35. Operating elements, such as spring loaded pins and movable arms, and visual elements, such

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as covers for light emitting diodes, shall be usable after applying the finish. The final film shall conform to color 34087 of FED-STD-595.

3.2.5 Marking. The unit shall be marked in accordance with MIL-M-13231. Panel marking shall be Group I as described in MIL-M-13231. The pin designations of the different voltages at the output connector shall be listed as a special characteristic.
* Both the Generator Head and Handle Assembly (11733321) and the
* Power Conditioning Unit (11733425) shall be marked with the unit serial number.

3.2.5.1 Serial numbers. The serial number of the Generator, Direct Current: G-76( )/G, shall be assigned in accordance with MIL-M-13231.

3.2.5.2 Visibility. Wherever practicable, components and assemblies shall be mounted so that their identification markings are readily visible with minimum disassembly of equipment.

3.3 Preconditioning.

3.3.1 Burn-in. Prior to delivery and after all processing,
* each Generator Head Assembly (11733284) and Power Conditioning
* Unit (11733285) shall be burned in (See 4.8.1).

* 3.3.2 Bounce preconditioning. The Generator Head Assembly
* (11733284) and Power Conditioning Unit (11733285) shall meet requirements (3.4) after being subjected to bounce preconditioning (See 4.8.2).

3.4 Performance requirements. The performance requirements contained herein are mandatory.

3.4.1 Rated output (resistive load). The unit (pins B and D of output connector) shall provide a current of not less than 2 amperes into a resistive load of 10 ohms with an output voltage not less than 20 volts at a cranking speed of 68 rpm; pins A and C are open circuited (See 4.8.3).

* 3.4.2 Output voltages. Tests to be performed at ambient
* temperature unless otherwise specified.

* 3.4.2.1 Regulated output. The regulated output voltage (pins
* B and D) of the unit while cranking at 105 rpm into a 100 ohm
* load shall not be less than 30 volts and shall not exceed 32.0
* volts DC at ambient temperature and shall not be less than 29.6
* volts at -10°F nor more than 32.5 volts at 120°F (see 4.8.4).

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3.4.2.2 Unregulated output. The unregulated output voltage * (pins A and C) with a 10-ohm load shall be no less than 20 VDC at * a cranking speed of 100 rpm (see 4.8.4).

3.4.3 Heavy load. The unit shall withstand repeated heavy loads of not less than 5.0 amperes for one minute across pins B and D, pins A and C being open circuited, with cranking speeds of 100 rpm (See 4.8.5).

3.4.4 Indicators. A light emitting diode (LED) shall indicate when the unit is providing 24 volts ± 10 percent (See 4.8.4). A * second LED shall indicate when the unit is providing 2 amperes ± 10 percent (See 4.8.4 and 4.8.5).

* 3.4.5 Torque test. The unit shall require a torque of 170 * inch-pounds or less when cranking at 68 rpm into a resistive load * of 10 ohms connected across pins B and D of the output connector; * pins A and C being open circuited (see 4.8.6).

3.4.6 Protection. The unit shall be capable of withstanding repeated momentary short circuits at one-minute intervals on the regulated line (See 4.8.7).

3.4.7 Reverse current. The unit shall not discharge the battery to which it is connected for charging; reverse current at 50 volts shall not exceed 75 microamperes on either line (See 4.8.8).

3.5 Environmental requirements. The unit shall meet the * following environmental requirements. (Units shall be unpackaged * unless otherwise specified).

3.5.1 Temperature. Unit structural (3.2) and performance (3.4.1) requirements shall be met after storage at temperature * extremes of -50° to +160° F; performance requirements (3.4.1) * shall be met at temperatures of -50° to +120°F; and performance * requirements (3.4.2) shall be met at temperatures of -10° to * +120°F (See 4.8.9 through 4.8.10).

3.5.2 Vibration and shock. Unit and carrying bag structural (3.2) and performance (3.4) requirements shall be met after vibration and shock incident to field transport as loose cargo, rough handling in loading and unloading, and bench handling in servicing (See 4.8.11 through 4.8.16).

3.5.3 Humidity. Unit structural (3.2) and performance (3.4) requirements shall be met after prolonged exposure to humidity conditions encountered in tropical areas (See 4.8.17).

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3.5.4 Immersion. The unit shall meet performance requirements * (3.4) after being immersed in water for 10 minutes. There shall * be no bubbles coming from within the equipment (see 4.8.18).

3.5.5 Salt atmosphere. Unit structural (3.2) and performance (3.4) requirements shall be met after exposure of the unpackaged unit to salt atmospheres as may be encountered in coastal areas (See 4.8.19).

3.5.6 Fungus. Unit structural (3.2) and performance (3.4) requirements shall be met after prolonged exposure of the unpackaged unit to conditions favorable to fungus attack, as may be encountered in tropical climates (See 4.8.20). Examination after exposure shall show no evidence of fungus.

3.5.7 Blowing sand and dust. The unit shall meet performance requirements (3.4) after being subjected to blowing sand and dust as encountered in the desert (See 4.8.21).

3.6 Electromagnetic compatibility. The unit shall comply with the radiated and conducted emission requirements of MIL-STD-461, * Notice 4, as specified below:

RE 02
* RE 02.1
CE 04

(See 4.8.22.)

3.7 Reliability. The unit shall have a mean time between failure (MTBF) of 750 hours minimum acceptable value (MAV) and a best operation characteristic (BOC) of 2250 hours (a discrimination ratio of 3). The requirement applies under the following conditions:

a. Temperature range of 95° to 113° F.
b. Continuous operation with air cooling of generator bearings and regulator box.
c. Unless otherwise specified, the environment will be ground mobile as defined in MIL-STD-781.

(See 4.8.23.)

3.8 Workmanship. The equipment shall be manufactured and assembled with the applicable portion of the following paragraphs of MIL-P-11268.

a. General requirements for plastic material and plastic parts.

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b. Wiring and cabling, including: slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

c. Flux and cleaning agents for solder.

d. Process for solder.

e. Cleaning

f. Riveting

g. General requirements for securing parts.

h. Welding

i. Mechanical operation of controls

j. Self-tapping screws

Requirements of MIL-STD-454, Requirement 9, shall be used to establish acceptable workmanship criteria.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all required inspections as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspections specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established by the contractor and approved by the Government prior to use. Gage and instrument settings shall be a function of the equipment and independent of the operator. Test equipment shall be marked to indicate the * limit of acceptance for each test. Test equipment accuracy shall be taken into account in setting the acceptance limits. Special test equipment required for acceptance tests is listed in table I and EDL 11737594.

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Table I. Special Inspection Equipment (SIE)

Reference	Functional Description	Dwg. No.
4.8.1	Tester with motor to drive generator at 68 rpm	11734018
4.6.1 & 4.8.3 through 4.8.7	Tester with motor to drive generator at required speeds of 0 to 100 rpm and with instruments to monitor electrical performance and mechanical energy applied	11734019
4.6.1.2.2 & .3	Test circuit for A & B Boards	11734020

* 4.1.2 Standard conditions. Tests shall be conducted at standard ambient conditions of temperature, pressure, and humidity as defined in MIL-STD-810, unless otherwise specified. The following tolerances on test conditions also apply, unless otherwise specified:

a.	Time	±2 percent
* b.	Resistance, current	±2 percent
c.	Frequency, period	±2 percent
d.	Vibration amplitude	±10 percent
e.	Force, weight, or torque	±2 percent
* f.	Voltage	±0.5 percent

*

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (does not include packaging). (See 4.3)

b. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.4)
(2) Quality conformance inspection of packaging. (See 4.9)

c. Inspection covered by subsidiary documents. (See 4.5)

4.3 First article inspection. The contractor shall inspect attributes of ten sets of parts listed in 6.4. The measurements shall be compared with the drawings, and the comparison shall be submitted with the first article sample test data. In addition

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the data on the ten sets of parts shall be certified by an independent laboratory approved by the Government. The first article sample as specified in 3.1.1 shall be inspected as specified in table II, Classification of Defects (4.6), and Material and Component Certifications(4.7). The tests of samples shall be conducted in the order specified in the table. The tests shall be performed by the contractor under the observation of the Government QA representative and/or a representative of the cognizant technical agency.

4.3.1 Approval of first article sample. If the sample passes the criteria of specified examinations and tests, it shall be approved. If the sample fails in any of the examinations and tests, failure analysis shall be conducted by the contractor under Government surveillance. The results of the tests and failure analysis of the units, together with the engineering analysis of the units, shall form a basis for corrective action. Depending upon the degree of corrective action deemed necessary by the Government, the FAS may be:

Table II. First Article Sample Inspection.

Inspections	Req Para	Test Para	Unit Number
Inspection Covered by Subsidiary Documents	-	4.5	All
All Group A tests	See Table III		All
Group B tests*	See Table IV		All*
High Temperature	3.5.1	4.8.9	1,2
Low Temperature	3.5.1	4.8.10	1,2
Vibration	3.5.2	4.8.11	1,2
Vibration-Temperature	3.5.2	4.8.12	3,4
Shock (Drop)	3.5.2	4.8.13	1,2
Shock (Drop-Temperature)	3.5.2	4.8.14	3,4
Bounce	3.5.2	4.8.15	1,2
Shock (Bench)	3.5.2	4.8.16	1,2
Humidity	3.5.3	4.8.17	5
Immersion	3.5.4	4.8.18	6
Salt fog	3.5.5	4.8.19	7
Fungus	3.5.6	4.8.20	8
Blowing Sand and Dust	3.5.7	4.8.21	9
Electromagnetic compat	3.6	4.8.22	10
Reliability	3.7	4.8.23	11 - 20

*All units shall be subjected to all Group B tests except the temperature tests, which will be conducted as indicated in this table; AQL's in table IV do not apply.

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a. Accepted, provided recommended changes are made by the contractor prior to the start of regular production or during production of the first lot as prescribed by the Government. In the latter case, such portions of the first article inspections needed to verify corrective action may be included in the acceptance requirements of the lot at the Government's discretion.

b. Rejected, and new samples required for the tests failed and for related tests which might be invalidated by the corrective action.

c. Rejected, and a complete new sample required for approval.

In all cases, the contractor shall comply with any required changes to the satisfaction of the Government for the duration of the contract.

4.3.2 Disposition of first article sample. The ten sets of parts on which attributes measurements were made shall be among those used in fabricating the twenty FAS units. After inspection the FAS units shall be washed, cleaned, dried, and refurbished as * necessary and shall be submitted with a subsequent lot, which will be subjected to quality conformance tests.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract or purchase order. The contractor shall supply complete data on the contractor tests and examinations of the FAS units in a Government approved reproducible form.

4.3.4 Reinstitution of tests. Acceptance of the FAS does not relieve the contractor from meeting all requirements of this specification throughout the contract. The Government reserves the right to independently verify all requirements by repeating FAS inspections or any portion thereof on any production lot. If such verification indicates failure to meet requirements previously verified by FAS tests, the contractor shall be required to take corrective action. If the requirements are not met on the first lot to which the correction can be applied, the Government reserves the right to institute tightened inspection on a lot basis, by adding to the normal quality conformance inspections all or such parts of the FAS tests needed to verify corrective action. Normal quality conformance inspection will be resumed after two consecutive lots have satisfactorily passed the tightened conditions.

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4.4 Quality conformance inspection of equipment before packaging. Inspection shall be as specified in this document. The contractor shall notify the Government QA representative prior to proceeding with inspection for Classification of Defects (4.6) and Group A and B tests (Table III and IV). Inspection may be made at a higher assembly level than specified provided that equal quality assurance provisions are satisfied, but not at a lower level than specified.

4.4.1 Lot formation. A lot shall consist of units produced at one manufacturing location in accordance with stabilized methods and techniques, the same drawing, and specification, and revisions (except for those introduced without obsolescence), under continuous production. One lot shall comprise one month's production but not less than 50 units.

4.4.2 Sampling. Sampling shall be in accordance with MIL-STD-105 and this document. Unless otherwise specified, inspection level II shall apply.

4.4.3 Lot acceptance. Lots shall be accepted on the basis of inspection by subsidiary documents (4.5), Classification of Defects (4.6), Material and Component Certification (4.7), Group A Tests (Table III), and Group B Tests (Table IV). If a unit should fail any Group B test, testing will be stopped and data and analysis will be forwarded to the technical agency within two working days. The technical agency will analyze the failure and recommend changes where indicated.

4.4.4 Data recording. A record shall be made of the results of all examinations and tests performed as specified elsewhere in section 4 of this specification. The test data shall be recorded on computer compatible input medium that is preferred and approved by the Government. Data from each performance test shall be recorded, but only the defect summary need be recorded in the inspection of the Classification of Defects. Data layout shall be approved by the Government. The data shall be forwarded to the designated Government agency within two working days after the Government acceptance of the item or lot. If inspections are conducted on a sublot basis, the acceptance data for each sublot shall be so identified. (See Engineering information, Section 6.3).

4.4.5 Disposition of lot samples. After passing the prescribed tests, the samples shall be refurbished as necessary. One sample (comparison sample) shall be sent to the technical agency (see 6.1.3); the remaining samples shall be returned to the lot.

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engine shall be started with the exhaust line closed [illegible] tube. The

Table III. Group A tests

Inspections	Reqt Para	Test Para	Test Equip	Acceptance Criteria
Burn in	3.3.1	4.8.1	SIE	100% screening
Bounce preconditioning	3.3.2	4.8.2	STE	100% screening
Rated output	3.4.1	4.8.3	SIE	100% Screening
Output voltage	3.4.2	4.8.4	SIE	100% Screening
Heavy load	3.4.3	4.8.5	SIE	100% Screening
* Indicators, 24 volt	3.4.4	4.8.4	SIE	100% Screening
* Indicators, 2 amperes	3.4.4	4.8.5	SIE	100% Screening
Torque	3.4.5	4.8.6	SIE	100% Screening

Table IV. Group B tests

Inspections	Reqt Para	Test Para	Test Equip	Acceptance Criteria
Protection	3.4.6	4.8.7	STE	AQL 1.0
Reverse current	3.4.7	4.8.8	STE	AQL 1.0
High temperature	3.5.1	4.8.9	STE	AQL 1.0
Low temperature	3.5.1	4.8.10	STE	AQL 1.0

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract or purchase order.

Item	Where Required
Construction	3.2
Materials	3.2.1
Accessories	3.2.2
Dimensions	3.2.3.1
Finish	3.2.4
Marking	3.2.5
Workmanship	3.8
Preservation-Packaging	5

4.6 Classification of defects. The assembly shall be inspected by using MIL-STD-252 and the following Classification of Defects and applicable drawings.

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4.6.1 Hand crank generator assembly (11733326).

Category	Defect	Method	Acceptance Criteria
Major			
101	Running torque	DS-1 and SME	100% Inspection
Minor			
201	Torque on all fasteners	SME	AQL 1.0
202	Function of hinges	Manual	AQL 1.0
203	All weld joints	Visual/SME	100% Inspection
204	Insure operation of proper locking mechanism	Manual	100% Inspection
205	Finish	Visual	100% Inspection

* 4.6.1.1 Generator head and handle assembly (11733321).

Category	Defect	Method	Acceptance Criteria
Minor			
201	Movement of hinged parts	Manual	100% Inspection
202	Presence of internal tooth lock washer and spacer washer	Visual	100% Inspection
203	Proper torque on all screws	SME	AQL 1.0

* 4.6.1.1.1 Generator head assembly (11733284).

Category	Defect	Method	Acceptance Criteria
Major			
101	Water seal	DS-2	2 samples/lot ac 0/re 1
Minor			
201	Conformal adhesive	Visual	100% Inspection
202	Torque on screws	SME	AQL 1.0

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4.6.1.1.2 Electromechanical assembly (11733285).

Category	Defect	Method	Acceptance Criteria
Minor			
201	Adhesive on screws	Visual/SME Torque Test	AQL 1.0
202	Proper position of low speed shaft bearings	DS-3	100% Inspection
203	Cleanliness of harmonic drive	Visual	100% Inspection
204	Lubrication of harmonic drive	Visual (11734028)	100% Inspection

4.6.1.1.3 Alternator cup and high speed shaft assembly (11733314).

Category	Defect	Method	Acceptance Criteria
Minor			
201	Proper weight, amount, and distribution of grease	(11734028) Visual	100% Inspection
202	Torque on screws	SME	AQL 1.0

4.6.1.1.4 High speed shaft assembly (11733288).

Category	Defect	Method	Acceptance Criteria
Major			
101	Crimp rollover	DS-4	100% Inspection
Minor			
201	Torque on screws	SME	AQL 1.0

4.6.1.1.5 High speed shaft (11733303).

Category	Defect	Method	Acceptance Criteria
Minor			
201	Proper interference fit	SME	100% Inspection
202	Proper fit of woodruff key	Visual	100% Inspection

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* 4.6.1.2 Power conditioning unit (11733425).

Category	Defect	Method	Acceptance Criteria
Major			
101	Solder seal on plate screws	Visual	100% Inspection
102	Seal provided by plate and connector gaskets (water seal)	DS-2	2 samples/lot ac 0/re 1
Minor			
201	Heat sink and thermal adhesive	Visual	100% Inspection

4.6.1.2.1 Transformer (11733388).

Category	Defect	Method	Acceptance Criteria
Major			
101	Input and output DC resistance	SME	AQL 1.0
102	Input vs output AC voltage	SME	AQL 1.0
Minor			
201	Encapsulation and Condition	Visual	AQL 1.0
202	Overall maximum dimensional check	SME	AQL 1.0
203	Wire size and insulation	Gage	AQL 1.0

4.6.1.2.2 Board assembly A (11733389).

Category	Defect	Method	Acceptance Criteria
Major			
101	Electrical output	(11734020)	100% Inspection
102	Torque on heat sink screw	SME	100% Inspection
103	Part inventory	Visual	100% Inspection

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4.6.1.2.3 Board assembly B (11733411).

Category	Defect	Method	Acceptance Criteria
Major			
101	Electrical output	(11734020)	100% Inspection
102	Torque on heat sink screw	SME	100% Inspection
103	Part inventory	Visual	100% Inspection

4.6.1.2.4 LED support assembly (11733419)

Category	Defect	Method	Acceptance Criteria
Major			
101	Electrical function	STE	100% Inspection

4.6.1.3 Leg assembly, L.H. (11733337).

Category	Defect	Method	Acceptance Criteria
Minor			
201	Torque on bracket plate assembly and proper movement of plate	SME	AQL 1.0
202	Torque on pivot block leg	SME	AQL 1.0

4.6.1.4 Leg assembly, R.H. (11733339).

Category	Defect	Method	Acceptance Criteria
Minor			
201	Torque on bracket plate assembly and proper movement of plate	SME	AQL 1.0
202	Torque on pivot block leg	SME	AQL 1.0

4.6.2 Defect summary.

DS-1 For the proper torque measurement, refer to drawing 11734019.

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DS-2 Immerse in accordance with MIL-STD-810, Method 512.1, Procedure 1. Any evidence of moisture on disassembly shall constitute a failure.

DS-3 To check low speed shaft, fit bearing on both ends. Place TRUARC in shaft groove, place bearings in fully seated position, measure distance from outside to outside of bearing. This distance shall be as indicated on the drawing.

DS-4 Hold high speed shaft stationary, apply 50 inch-pounds of torque to bearing inner race. The bearing shall not move on the shaft.

4.7 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statements of findings that only materials and components conforming to the item specification, drawings, and subsidiary documents (4.5) have been used.

4.8 Test methods. (See 4.1.3)

* 4.8.1 Burn-in. The generator head assembly (11733284) with a
* power conditioning unit (11733425) attached shall be connected to an electric motor and be operated over a 3-hour period (one hour on, one hour off, one hour on) at a speed of 68 rpm. A suggested test setup is shown in 11734018.

4.8.2 Bounce preconditioning. The generator head assembly
* (11733284) with a power conditioning unit (11733425) attached
* shall be placed in their normal operating position on the table of a package tester (type 1000 SC, L.A.B. Corporation, Skaneatales, New York, or equivalent). The package tester, shafts in phase, shall have a speed such that the items bounce 1/4 inch from the bed. The items shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to being subjected to performance requirements.

* 4.8.3 Rated output. The generator head assembly (11733284)
* with a power conditioning unit (11733425) attached (pins B and D) shall be connected to a resistance load of 10.0 ohms and driven at 68 rpm. The output current shall be measured with an external ammeter and shall meet the output requirements of 3.4.1.

* 4.8.4 Output voltage. The generator head assembly (11733284)
* with a power conditioning unit (11733425) attached shall be
* driven at 105 rpm into a 100 ohm load. The regulated output
* voltage shall meet the requirements of 3.4.2.1. The accuracy of

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*the voltage indicating LED shall be determined by comparison with *an external voltmeter. Driving the generator head assembly and *power conditioning unit at 68 to 100 rpm into a 10 ohm load, the *accuracy of the LED shall meet the requirements of 3.4.4. The *voltage output of the unregulated line shall be measured while *cranking at 105 rpm and shall meet the unregulated requirements *of 3.4.2.2.

* 4.8.5 Heavy load. The generator head assembly (11733284) with *a power conditioning unit (11733425) attached shall be driven at 100 rpm, and the resistance load (across pins B and D) shall be adjusted to obtain an output current of at least 5.0 amperes. The generator assembly shall be so driven 10 times for 1.0-minute periods with 5-minute intervals between tests. The generator and conditioning unit shall meet the rated output requirements *(3.4.1) after this test. The accuracy of the 2.0-ampere indicating LED shall be determined by comparison with an external ammeter and shall meet requirements of 3.4.4.

* 4.8.6 Torque. The mechanical torque applied to the generator *assembly shall be measured using a suggested test set-up as shown *in HDL DWG 11734019. The unit shall meet the requirements of *3.4.5.

* 4.8.7 Protection. The generator head assembly (11733284) with *a power conditioning unit (11733425) attached shall be driven at 68 rpm, and a short circuit shall be momentarily applied across the regulated voltage terminals. The output shall be short circuited a minimum of five times with one-minute intervals between short circuits. During the short circuit, the cranking *speed shall be maintained. After the test, the generator head *and power conditioning unit shall meet the requirements of 3.4.2.1 and 3.4.6.

4.8.8 Reverse current. A DC power supply with an adjustable output voltage from 0 to 50 VDC shall be connected to the output terminals on both circuits of the generator assembly/conditioning unit in correct polarity with the generator not being cranked. The voltage of the power supply shall be gradually increased to 50 volts DC and requirements of 3.4.7 shall be met. The generator and conditioning unit shall meet the rated output requirements (3.4.1) after this test.

* 4.8.9 High temperature. The unit shall be temperature *conditioned in accordance with MIL-STD-810, Method 501.1, *Procedure II (See 3.5.1). At least once during each 12-hour *cycle after the unit has stabilized at 120°F, once at 120°F after *the third cycle, and once at ambient temperature upon completion

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*of the test, the unit shall undergo the rated output test (4.8.3) *and the output voltage test (4.8.4). The unit shall meet *requirements (3.4.1 and 3.4.2).

4.8.10 Low temperature. The unit shall be temperature conditioned in accordance with MIL-STD-810, Method 502.1, Procedure I (See 3.5.1). At least once during the -50°F portion of the test, after the unit has stabilized, and once at ambient temperature upon completion of the test, the unit shall undergo the rated output test (4.8.3). The unit shall meet the rated *output voltage requirement (3.4.1). At least once during the *-10°F portion of the test, the unit shall undergo the output *voltage test (4.8.4) and meet the requirements of 3.4.2.

4.8.11 Vibration test. The unit shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure X, curve AX (See *3.5.2). The generator head assembly (11733284) and power *conditioning unit (11733425) shall be tested and shall meet the *requirements of 3.4.1 and 3.4.2.

4.8.12 Vibration-temperature test. The unit in its combat carrying bag shall be tied down to the vibration fixture in accordance with the normally prescribed tie down procedures so that vibration is applied through the unit's vertical axis. The unit shall be at a temperature of 73°F. The unit shall be vibrated for two hours in accordance with the random vibration spectrum shown in figure 1. The inputs specified are those associated with the fixture and not the unit. The control accelerometer shall be on the fixture. Input tolerances for random vibration are to be held to ±3 dB up to 200 Hz and ±5 dB over 200 Hz. At completion, the unit shall be subjected to the following sinusoidal vibration environment:

Frequency (Hz)	Vibration Amplitude (g)	Sweep Rate	Total Time/Axis (min)
10 - 50	1.5	10-150 Hz	5
50-150	2.5	in 5 min	

The entire procedure described above shall be repeated two more times for a total of three cycles, first with the vibration imparted through the unit's longitudinal axis, and second with the vibration imparted through the unit's transverse axis. At the completion of each of the three cycles, the unit shall meet the requirements of paragraphs 3.5.2 and 3.4.1, and the combat carrying case shall meet the requirements of 3.5.2. The entire procedure described above shall be repeated two more times, once

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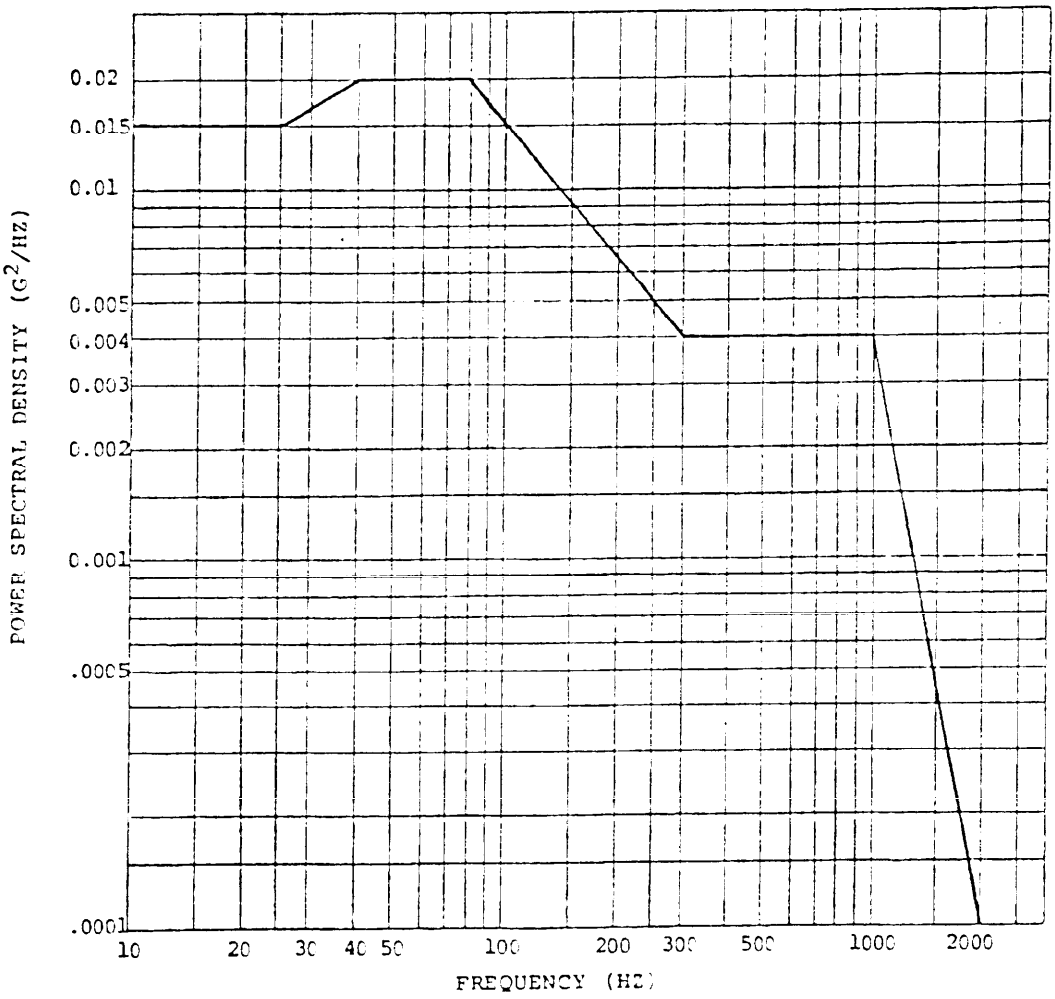


Figure 1. Surface and air transportation vibration.

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at a temperature of 130°±2.5°F and again at a temperature of -30°±2.5°F. These temperature extremes are to be achieved at a rate of 25° per hour, and the unit shall be stabilized at the test temperature for at least 16 hours before vibration. The control temperature sensor shall be attached to the skin of the container. During changes in vibration axes, the unit test temperature shall be maintained. The unit shall be returned to ambient temperature before the functional tests and appropriate inspections are conducted. At the completion of the high temperature testing, the unit shall meet the requirements of paragraphs 3.5.2 and 3.4.1; at the completion of the low temperature testing, the unit shall meet the requirements of 3.5.2 and 3.4.1.

4.8.13 Shock (Transit Drop) Test. The unit shall be conditioned in accordance with MIL-STD-810, Method 516.2, Procedure II (See 3.5.2). After conditioning, the unit shall meet requirements of 3.4.1 and 3.5.2.

4.8.14 Shock (Drop-Temperature) Test. The unit in its combat carrying bag shall be placed inside a conditioning chamber at ambient conditions. The chamber temperature shall be increased at the rate of 25°F per hour and stabilized at 130°±2.5°F for a period of not less than 16 hours. The unit shall be removed from the conditioning chamber and raised to a height such that its lowest corner is 30 inches above a concrete impact surface. The generator shall be the impacting part of the unit. Within five minutes after removal from the conditioning chamber, the unit shall be released using a quick-release mechanism. The unit shall be oriented at release such that its center of gravity is directly above the striking corner. After visually inspecting the unit and combat carrying bag for damage, it shall be returned to the conditioning chamber and reconditioned at 130°F±2.5°F for a period of not less than 1 hour and dropped on a different corner and inspected. This procedure will be followed until the unit has been dropped on all four of the generator's corners. At the completion of the four drops and inspections, the unit shall meet the requirements of paragraph 3.4.1 and 3.5.2. Using a second unit and combat carrying bag, the entire procedure described above shall be repeated but at a conditioning temperature of -30°F±2.5°F.

4.8.15 Bounce (loose cargo) test. The packaged unit shall be tested in accordance with MIL-STD-810, Method 514.2, Procedure XI, Part 2 (See 3.5.2). After conditioning, the unit shall meet requirements of 3.4.1 and 3.5.2.

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4.8.16 Shock (bench handling) test. The unit shall be tested in accordance with MIL-STD-810, Method 516.2, Procedure V (See 3.5.2). The unit shall meet requirements of 3.4.1 and 3.5.2 after conditioning.

4.8.17 Humidity test. The unit shall be tested in accordance with MIL-STD-810, Method 507.1, Procedure III and shall meet the requirements of 3.4.1 and 3.5.3 during and after conditioning.

4.8.18 Immersion test. The unit shall be tested in accordance * with MIL-STD-810, Method 512.1, Procedure IV. The equipment * shall be pressurized to 4 psig for a minimum of 10 minutes. The * unit shall meet the requirements of 3.4.1 and 3.5.4 after * conditioning before opening the unit.

4.8.19 Salt fog test. The unit shall be tested in accordance with MIL-STD-810, Method 509.1 Procedure I. After conditioning, visual inspection shall show no flaked, pitted, blistered, or otherwise loosened finish or metal surface. The unit shall meet requirements of 3.4.1.

4.8.20 Fungus test. The unit shall be tested in accordance with MIL-STD-810, Method 508.1. On completion of conditioning, the unit shall be inspected in accordance with Procedure I, and shall meet requirements of 3.5.6 and 3.4.1.

4.8.21 Blowing sand and dust test. The unit shall be subjected to tests in accordance with Method 510.1, Procedure I, MIL-STD-810 and shall meet requirements of 3.4.1 and 3.5.7 after the test. Dust in any component shall be cause for failure.

4.8.22 Electromagnetic compatibility. The unit shall be subjected to an electromagnetic interference test in accordance * with MIL-STD-462, Notice 4, Methods RE 02, RE 02.1 and CE 04 and * shall meet requirements of 3.6.4.

4.8.23 Reliability test. The reliability test shall demonstrate the MTBF required in paragraph 3.7. The test shall be run according to the conditions of MIL-STD-781, test plan XXI-* C. The generator head assembly (11733284) and power conditioning * unit (11733425) shall be connected to a resistance load of 10.0 ohms and cranked at a speed of 68 rpm. A failure for reliability purposes shall be defined as the failure of any part of the unit to meet the requirements of this specification. Each unit shall be continuously monitored to detect any failures.

4.9 Quality conformance inspection of packaging. Packaging shall be inspected as provided by SPI 1G0061 to determine conformance with the requirements.

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4.9.1 Rough handling test (packaging). When the rough handling test in accordance with MIL-P-116 is required by contract the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling:

Inspection	Measurement	Cranking Speed
3.4.1	4.8.3	68 rpm

5. PRESERVATION-PACKAGING

5.1 Preservation-packing. Preservation and packing shall be in accordance with SPI 1G0061.

5.2 Shipping containers. Shipping containers shall be as specified in SPI 1G0061.

* 5.3 Marking.

5.3.1 Military. In addition to any special marking required by the contract or order, interior packs and exterior shipping
* containers shall be marked in accordance with 3.2.5 and MIL-STD-129.

6. NOTES

6.1 Item definition. The Generator, Direct Current, G-76( )/G, shall meet the requirements of HDL drawing number 11733326.

6.1.1 Interface definition. The G-76( )/G shall meet the interface requirements specified in HDL drawing number (11733426).

6.1.2 Additional items. A combat carrying bag, cable assemblies or angled coupler, and technical literature will be provided as specified in the contract and be packed with the generator as specified in the contract. (See 6.2.)

6.1.3 Government testing of comparison samples. These samples are subject to periodic retesting by the Government to insure that first article approval requirements continue to be met. Failure of these samples will not affect the acceptance of the lot, but may indicate future mandatory action.

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6.1.4 Lead time. Government testing or analysis of the first article approval sample will require a maximum of 20 working days. If there are failures, additional time will be required.

6.2 Ordering information.

6.2.1 Documents. Procurement documents should specify the title, date, and number of this specification.

6.2.2 First article. As specified by the contract, if a first article sample is required, the tested units may be refurbished and added to the quantity required for delivery.

6.2.3 Data requirement. The following data are required by this specification to be furnished to the Government and should
* be entered on DD Form 1423 per Data Item Descriptions indicated
* if made a part of the contract: First article inspection results
* (DI-T-4902/M1), lot acceptance inspection results (DI-P-1757),
* contractor furnished inspection equipment designs (DI-R-1713).

6.2.4 Level of packaging. The contract should specify Level A or B preservation and packing (See section 5). In addition to
* the generator, the carrying bag and technical literature, the normal pack shall include:

Part No.	Nomenclature
* 11733296	Cable Assembly, Special Purpose, Electrical, CX-13175/G
* 11733297	Cable Assembly, Special Purpose, Electrical, CX-13176/G

* As an option, the pack may include only Part Number 11733985,
* Adapter, Connector, MX-2122/G and Part Number 11738274, Cover,
* Plate, Nonskid, Assembly, instead of cables.

6.3 Engineering information. Upon request, copies of the data collected in the acceptance of a lot will be made available to the Government representative. Additional data or data requiring special requirements, when needed for engineering purposes, will be specifically defined by the contract.

6.3.1 Harmonic drives. The harmonic drives used in the G-76( )/G shall be interchangeable with one another. The components of any one harmonic drive shall not necessarily be interchangeable with the components of another harmonic drive.

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6.4 First article sample parts list. Ten sets of the following parts shall be compared with drawings as specified.

Drawing No.	Title	Characteristic
* 11733332	Seat	All physical
* 11733311	Block, Hinge	All physical
* 11738248	Shaft	All physical
* 11738249	Hinge, Block	All physical
* 11733356	Pivot Block, Leg	All physical
* 11733338	Leg	All physical
* 11733375	Sleeve, Rotor	All physical
* 11733374	Magnet	All magnetic
* 11733373	Core Rotor	All physical
* 11733353	Rotor Assembly	All physical
* 11733303	Shaft, High Speed	All physical
* 11733377	Lamination Stack	All physical
* 11733302	Cup, Alternator	All physical
* 11733291	Spacer	All physical
* 11733352	Shaft, Low Speed	All physical
* 11738236	Arm, Crank, Right	All physical
* 11738239	Arm, Crank, Left	All physical
* 11733290	Cap, End, No. 1	All physical
* 11733300	Cap, End, No. 2	All physical

* 6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER (Number) AND TITLE

MIL-G-49369A(ER) GENERATOR, DIRECT CURRENT: G-76( )/G with Accessories

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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